Enhancing Skin Disease Diagnosis with Convolutional Neural Networks

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***Abstract*— Skin diseases encompass a wide range of conditions, from acne and eczema to serious diseases like carcinoma. This study explores automated detection of these conditions using Convolutional Neural Networks (CNNs) by applying various architectures, including ResNet50, Xception, MobileNet etc. to two datasets: the "Augmented Skin Conditions Image Dataset" and the "Skin Diseases Dataset". In the Augmented Skin Condition Dataset, ResNet50 achieved the highest accuracy of 96.4%, indicating its effectiveness in skin condition classification. For the Skin Disease Dataset, Xception Net delivered the best accuracy at 95.73%, demonstrating its capability in detecting specific skin diseases. The results highlight the potential of CNNs for enhancing diagnostic accuracy across different datasets and underline the advantages of particular models for distinct classification tasks. This work advocates the use of light weight CNN models for detection of medical conditions in skin for early intervention of diseases and may be used for development of CAD systems.**

Keywords—Skin Condition Detection, Convolutional Neural Networks (CNN), Deep Learning, Medical Diagnostics, CAD systems.

I. INTRODUCTION

The skin is the body's largest organ, protecting it from outside elements while also controlling temperature, keeping moisture, and sensing exterior stimuli [1]. The epidermis, dermis, and subcutaneous tissue are among the layers that make up the skin, and it contains a wide range of cells and structures that work together to preserve health and aid healing. However, due to its frequent contact to numerous external substances, the skin is especially vulnerable to a variety of disorders, ranging from innocuous rashes to potentially fatal cancers such as melanoma.

Millions of individuals of all ages are affected by skin illnesses, which are among the most prevalent health problems in the world [2]. Effective treatment of skin conditions depends on early and accurate diagnosis, which also significantly lowers the risk of negative outcomes. However, some dermatological knowledge is often required for the diagnosis of skin problems, and this knowledge may not be easily accessible in all healthcare settings. The lack of resources, combined with the subjective nature of visual diagnosis, can result in misdiagnosis or delay in treatment. As a result, developing automated systems for accurately detecting skin conditions is critical to improving healthcare accessibility and results.

New developments in deep learning have paved the way for substantial progress in medical image analysis. One kind of deep learning is Convolutional Neural Networks (CNNs) [3]. architecture, have shown to be especially successful in image classification tasks. CNNs are capable of learning on their own, complex patterns and characteristics of pictures, which makes them ideal for use in medicine, such as skin disease detection. CNNs can interpret raw picture data directly, in contrast to standard machine learning models that depend on manually created features, enabling more accurate and efficient classification.

In recent years, methods like deep learning and machine learning have significantly advanced medical science. Today, machine learning is widely used in the medical field for tasks like virus detection [4], cancer prediction [5], osteoporosis prediction [6] etc. Specifically, Convolutional Neural Networks (CNNs) have become a powerful tool for image classification tasks, including medical image analysis, enabling accurate performance in assignments like malaria disease detection [7].

This paper aims to leverage CNNs, or convolutional neural networks, are used to categorize a range of skin conditions and diseases from two different skin condition datasets. By incorporating advanced architectures like VGG16 and ResNet50, the research seeks to enhance diagnostic accuracy and reliability in skin condition detection. The ultimate goal is to support healthcare professionals in making faster and more precise diagnoses, therefore enhancing patient results and contributing to the advancement of AI applications in dermatology.

The structure of this paper is organized as follows: Section II, "Related Work," summarizes findings from previous research. Section III, "Materials and Methods," describes the technologies employed and the approach taken in this study. Section IV, "Results and Discussion," presents the findings and their analysis. Finally, Section V, "Conclusion and Future Work," wraps up the paper and discusses potential directions for future research, followed by a References section.

II. RELATED WORK

This section provides the review of research various works performed in the field of Skin Condition Detection. M. Abbas et al. [8] (2024) employed Convolutional Neural Networks (CNNs) to categorize skin disorders, combining machine learning methods with image processing techniques. The study's primary goal was to improve the accuracy of skin disease identification while reducing false positives in the HAM10000 dataset. In order to tackle the problem of class disparity among the various skin disease categories, picture enhancement techniques were used. The study developed a sequential CNN model with with a 98% accuracy rate and an

Area Under the Curve (AUC) of 99% in each of the seven disease categories. The DenseNet-121 and ResNet-50 models had 89% and 84% accuracy rates, respectively. R. Sadik et al. [9] (2023) proposed an approach for detecting skin disorders that blends CNNs, MobileNet, and Xception architectures. To increase feature extraction, this method employs transfer learning by pre-training a model on the ImageNet dataset. The authors also evaluated the performance of many well-known deep learning models, including DenseNet, InceptionV3, ResNet50, and Inception-ResNet. They gathered information on five different skin conditions from two distinct sources. Several performance metrics, such as F1 score, precision, and recall, were employed to assess the models. The experimental results show that this strategy is effective, with the MobileNet attaining 96.00% classification accuracy and the Xception model, which uses transfer learning, achieving 97.00% accuracy.

O. Jaiyeoba et al. [10] (20024) did a study to increase the precision of skin disease classification using an ensemble method that used five algorithms. The study's purpose was to create a model for ensemble machine learning specifically for identifying Erythemato-Squamous Diseases (ESD) in order to deliver accurate artificial intelligence diagnosis and appropriate therapy. During training, the ensemble approach leverages the predictions of five classifiers (Naïve Bayes, Support Vector Classifier, Decision Tree, Random Forest, and Gradient Boosting) as input features for a meta-classifier. Each classifier obtained varied levels of accuracy, including Random Forest (97.91%), Decision Tree (95.13%), Support Vector Machine (98.61%), Naïve Bayes (85.41%), and Gradient Boosting (95.83%). Using the stacking strategy, the ensemble model surpassed the base models, obtaining an overall accuracy of 99.30%. Similarly, S. Inthiyaz et al. [11] (2024) proposed A skin disease diagnosis system based on image processing. The proposed study developed a technique for identifying different types of these disorders. The system analyzes the user's submitted image of the skin condition, uses the CNN algorithm to extract features, and uses the softmax image classifier to diagnose the condition. Consequently, a new dermoscopy classification and detection technique using CNN was developed. They argued that the methodology described in the study is better suited to detecting and diagnosing skin issues than current methods. The suggested Work aids in obtaining the best characteristics from the skin photos, and the Softmax classifier is then used to classify them, which is highly accurate and achieved an accuracy of 87%.

In their work, V. Gautam et al [12] (2023) conducted research with the primary purpose of classifying, detecting, and providing accurate information regarding skin illnesses. Their paper addressed the same problem by implementing a Convolution neural network (CNN) with good performance for early skin disease classification and detection. The study used 2000 skin disease images from a standardized dataset Dermnet. The pre-processed photos were examined at various stages with a Deep Convolution Neural Network (DCNN). The approach described in their research obtained an accuracy of 98% across multiple illness categories. Also, in their study, R. Sulthana et al. [13] (2024), presented a comprehensive methodology for identifying skin lesions. Their research identified their deep learning-based S-MobileNet model as the best solution for categorizing skin lesion photos in the HAM10000 dataset. The results showed that using a preprocessing strategy considerably enhanced model performance. A modified version of the Gaussian filtering algorithm was used to segment the images and extract features as well as SFTA. After processing, the entered the S-MobileNet model with the dataset, which achieved an accuracy of 98.345% via segmentation and feature extraction.

N. Kashyap et al. [14] (2024) created a convolutional neural network model with many classes to distinguish between healthy and sick skin.To properly address the issue, they developed a program that combined this convolutional neural network using the VGG19 and AlexNet pre-trained networks. They created their dataset by collecting pictures from two different sources: DermNet and HAM10000. They only included five different skin disorders in their model: Bullous pemphigoid, melanoma, eczema, nail fungus, and vascular tumor, with respective accuracy levels of 90.27%, 86.75%, 96.83%, 85.64%, and 84.22%. Similarly, In their paper, S. Ahmed et al. [15] (2023) proposed a deep learning-based CNN technique for skin disease classification. They extracted around 2102 picture files from a huge archive of varied source photos, focusing on dermatoscopic images of pigmented lesions. Their findings indicate an 89.45% accuracy in diagnosing skin cancer illness. They used CNN algorithms to create a model for forecasting skin problems. Combining features using deep learning has been found to improve precision and forecast a much broader variety of diseases than previous models. They obtained an accuracy of 89.45%. Table I presents a summary of the literature evaluated in this section.

TABLE I. REVIEW OF LITERATURE

Year/Ref no.	Method/Model	Dataset Used	Performance
2024 [8]	CNN models-DenseNe-t121 and ResNet-50	More than 10,000 images representing seven different skin conditions	Accuracy: DenseNet-121-89%, ResNet-50- 84%
2023 [9]	CNN-MobileNet, Xception, ResNet50, InceptionV3, InceptionResNet, DenseNet	HAM10000 dataset	Accuracy-97.00%
2024 [10]	Stacking Ensemble Machine Learning (ML) Techniques	Skin diseases from University of California, Irvine (UCI) repository.	Precision-99.3%
2024 [11]	CNN- SoftMax classifier	Pretrained model of skin diseases from Kaggle	Accuracy- 87%
2023 [12]	Deep CNN	2000 skin disease pictures collected from a standardized dataset DermNet	Accuracy-98%
2024 [13]	Customized S-MobileNet CNN model	HAM10000 dataset	Accuracy-98.345%
2024 [14]	CNN- AlexNet and VGG19	DermNet and HAM10000	Accuracy-96.83%
2023 [15]	CNN algorithms	2102 image skin disease dataset	Accuracy-89.45%

III. MATERIALS AND METHODS

The subsequent sections extensively emphasize the diversity of resources used and provide detailed descriptions of the methodologies and procedures used in the production of this research project. The sub-sections are : A. Dataset Used, B. Data Augmentation, C. Technology Used and D. Methodology

A. Dataset Used

The first dataset utilized in this research was sourced from Kaggle. and is a real-world Augmented Skin Conditions Image Dataset [16]. The Augmented Skin Conditions Image Dataset comprised 2,394 augmented images representing six distinct dermatological conditions, making it a perfect tool for machine learning model training in medical image analysis. Each category contained 399 images, ensuring equal representation across all conditions. The dataset included images of acne, carcinoma, Eczema**,** Keratosis, Milia, and Rosacea. All images were in JPEG format with variable sizes.

The second utilized in this study was also obtained from Kaggle and is a real-world Skin Diseases Image Dataset [17]. The Skin Disease Dataset comprised images of eight types of skin infections collected from the internet,including parasitic (cutaneous larva migrans), fungal (athlete's foot, nail fungus, ringworm), bacterial (cellulitis, impetigo), and viral (chickenpox, shingles) illnesses. All images were in JPEG format with variable sizes.

B. Data Augmentation

Data augmentation is a crucial technique employed to increase the training dataset's diversity and enhance the generalization capability of deep learning models in this study. For the first dataset, Various augmentation strategies were implemented across different architectures, including EfficientNetB0, VGG16, and ResNet50. For the EfficientNetB0 model, transformations included random horizontal flips, slight rotations (up to 10 degrees), and random zooms (up to 10%), enabling the model to learn invariance to orientation and scale variations in skin lesions. Similarly, the VGG16 model utilized horizontal flips, random cropping and resizing, and color jittering to augment the dataset, promoting robustness against lighting variations and enablingthe model to concentrate on various parts of skin lesions. The ResNet50 model employed horizontal flips, random rotations, and random zooms, mirroring the techniques used in the other models.

Similarly, for the second dataset, a variety of augmentation techniques were applied like- Shear transformations, rotation, height and width changes, zooming, and horizontal flipping. Collectively, these data augmentation techniques enriched the training process, leading to improved classification performance and enhanced diagnostic accuracy for various skin conditions in clinical applications.

C. Technology Used

One kind of deep learning model designed especially to handle structured grid data, like photos, is called a convolutional neural network (CNN). CNNs, which are motivated by the visual cortex's biological functions, use a sequence of convolutional layers to automatically and adaptively learn spatial feature hierarchies from input photos.

EfficientNetB3 is an architecture for a convolutional neural network that is included in the EfficientNet family, which was introduced by Google in 2019. The EfficientNet models are designed to optimize both the accuracy and efficiency of deep learning models by scaling the network in a balanced manner. Its compound scaling approach allows it to achieve excellent precision while keeping the model size minimal, making it appropriate for use in a range of applications [18]. EfficientNetB3 is widely used in numerous computer vision tasks, including as segmentation, object detection, and picture categorization.

ResNet50 is a convolutional neural network architecture created by Microsoft Research in 2015. It is notable for its innovative application of residual connections, which solve the vanishing gradient issue that deep networks frequently face. It has 50 layers and enables for the training of very deep architectures using residual mappings rather than direct outputs, considerably simplifying the learning process. ResNet50 has demonstrated cutting-edge performance on benchmarks such as the ImageNet dataset, making it appropriate for a variety of applications, such as segmentation, object recognition, and image classification [19]. It typically operates on input images of size 224x224 pixels and is widely used for transfer learning, allowing practitioners to fine-tune the model on individual datasets by substituting the final classification layer.

The Inception architecture in Convolutional Neural Networks (CNNs) is intended to increase network efficiency by allowing the model to learn from multiple levels of feature representations at once [20]. Inception networks are known for their efficiency and depth, achieving excellent accuracy while reducing the number of parameters.

A deep learning architecture called DenseNet, sometimes referred to as Dense Convolutional Network, increases model efficiency and feature reuse by linking each layer directly to the next layer in a feed-forward fashion [21]. Unlike standard CNN designs, which simply transfer output to the next layer, DenseNet enables each layer to transfer feature mappings to each layer after it and accept input from all layers before it. This design allows the network to better use characteristics learned in previous layers, reducing the risk of vanishing gradients and boosting gradient flow.

The Visual Graphics Group (VGG) created the convolutional neural network architecture known as VGG16. This model is distinguished by its simplicity and depth, comprising 16 layers with learnable weights, hence the name VGG16. It uses small convolutional filters (3x3) stacked on top of each other to extract fine details and hierarchical information from images. VGG16 typically processes input images of 224x224 pixels and use max pooling layers to minimize spatial dimensions while keeping significant characteristics. The architecture has had a major influence on the deep learning sector, proving that increasing depthcan result in better performance on complicated tasks such as image classification [22], as proven by its success on the ImageNet dataset.

A lightweight and effective convolutional neural network (CNN) architecture, MobileNet was created especially for embedded and mobile vision applications [23]. Unlike traditional deep learning models that are computationally intensive, MobileNet emphasizes cutting down on complexity and size of the network without significantly compromising accuracy. It achieves this through two key innovations: depthwise separable convolutions and width multiplier. In many medical imaging projects, MobileNet has demonstrated its utility by providing high accuracy while requiring less computational power compared to larger models, making it perfect for deployment in settings with limited resources.

The Xception model, which represents "Extreme Inception," is a design for a convolutional neural network (CNN) that extends the success of the Inception model. Xception's main innovation is depth wise separable convolutions, which replace typical inception modules with a more efficient structure. In comparison to classic convolutions, our technique improves spatial feature learning while drastically lowering the quantity of parameters and computational requirements.

Convolutional neural networks (CNNs) with a deep architecture like VGG19 excels at picture categorization due to its simplicity and effectiveness. VGG19 is an expansion of the VGG16 model with 19 layers: 16 convolutional, three fully connected, and five max-pooling layers. VGG19's Convolutional layers employ simple 3x3 filters with padding and a stride of 1 allowing the model to catch minute details in input images while keeping the network manageable.

D. *Methodology*

The study employed a comprehensive methodology to classify skin conditions utilizing methods from deep learning, specifically Convolutional Neural Networks (CNNs). Techniques for augmenting data, including magnification, horizontal flipping, and random rotation, were used to enhance dataset diversity and improve model generalization. Six CNN architectures- ResNet50, VGG16, MobileNet, Inception, Densenet and EfficientNetB0- were implemented for the "Augmented Skin Conditions Image Dataset" and seven CNN architectures- EfficientNetB0, Xception, DenseNet121, InceptionV3, VGG16, VGG19 and MOBILENET- were implemented for the "Skin Diseases Dataset". The results showed different levels of accuracy among the CNN architectures, offering insights into the advantages and disadvantages of each strategy for classifying various skin conditions. Fig.1 represents the workflow of the study.

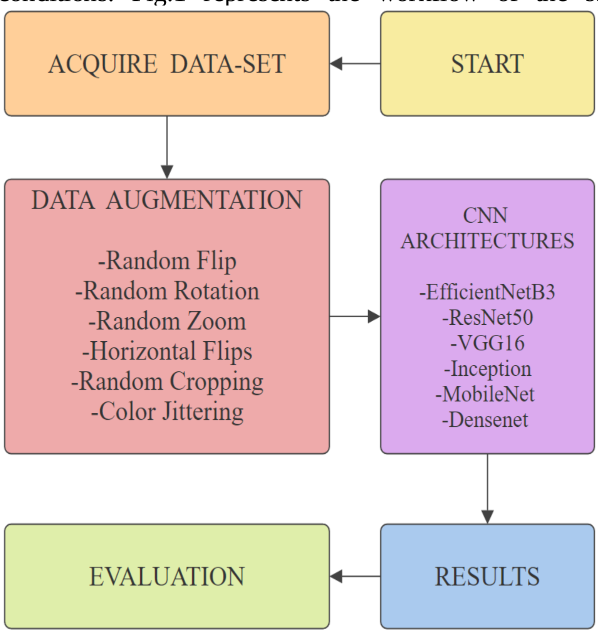

Fig. 1. Workflow Representation

IV. Results And Discussion

This section provides the results and the derived inferences which are briefly explained and analysed. All CNN architectures were developed and assessed using Python 3 with Anaconda Navigator for Jupyter notebooks. Following data augmentation, the previously mentioned CNN architectures were implemented. Evaluating the outcomes is critical to identify the optimal classifier. The primary performance metric used for evaluating the CNN architectures was Accuracy. Table II below shows the accuracy of the architectures in the first dataset i.e., "Augmented Skin Conditions Image Dataset" and graphically represented in Fig.2.

TABLE II. Architecture Performances

CNN Architectures	Accuracy
Inception	86.00
EfficientNetB3	92.80
ResNet50	96.40
VGG16	90.00
MobileNet	42.80
Densenet	70.00

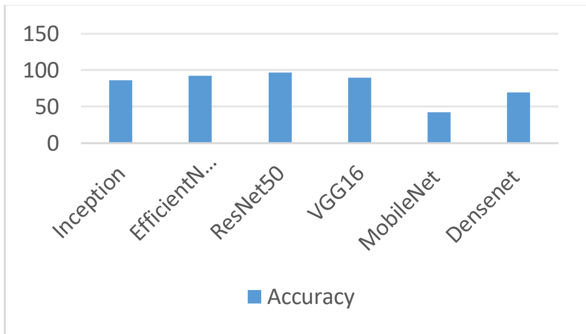

Fig. 2. Performance analysis of CNN models on dataset 1

For the first dataset, among the models tested, ResNet50 demonstrated the highest accuracy of 96.40%, indicating its effectiveness in distinguishing between the various dermatological conditions. EfficientNetB3 followed closely, obtaining an accuracy of 92.80%, demonstrating its proficiency in picture classification tasks. VGG16 scored a respectable 90.00% accuracy, indicating its potential for medical image analysis, albeit with somewhat lower performance than ResNet50 and EfficientNetB3.

Inception achieved an 86.0% accuracy, underscoring its capability in capturing multi-scale features. DenseNet achieved a moderate accuracy of 70.0%, suggesting its potential benefits of feature reuse but indicating room for improvement. MobileNet, however, underperformed relative to others, with a 42.8% accuracy, which may be attributed to its focus on model efficiency, potentially limiting its feature extraction capabilities on this dataset. These results emphasize the strengths and limitations of various CNN architectures for skin condition classification.

The contingency table, often known as the confusion matrix is a particular tabular depiction of a model's performance. In binary classification, it is made up of False positives (FP), false negatives (FN), true positives (TP), and true negatives (TN) are displayed in two rows and two columns. The confusion matrix shows that a great degree of accuracy is achieved by the ResNet50 model for most skin conditions, though some refinement is needed to better distinguish between visually similar classes such as Acne, Eczema, and Rosacea.

Fig.3 demonstrates the Confusion Matrix of the ResNet50 architecture. The confusion matrix shows that the ResNet50 model achieves a high level of accuracy for most skin conditions, though some refinement is needed to better distinguish between visually similar classes such as Acne, Eczema, and Rosacea. The ROC curves indicate that the ResNet50 model is highly effective at detecting different skin conditions, with particularly strong performance in detecting Carcinoma, Keratosis, Milia, and Rosacea. Acne and Eczema also show strong classification, though slightly less than the other conditions. Fig.4 Demonstrates the ROC curve for the ResNet50 architecture.

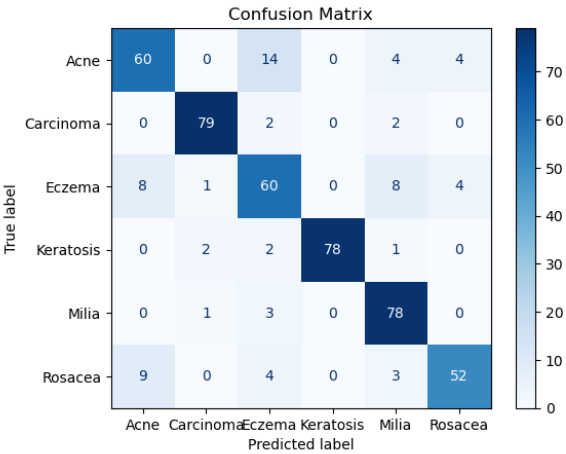

Fig. 3. Confusion Matrix of ResNet50

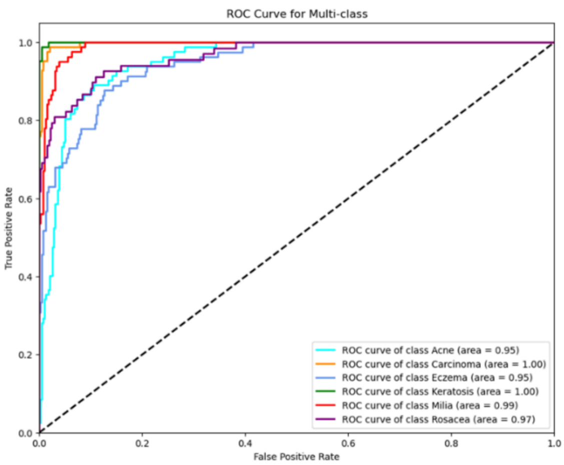

Fig. 4. ROC curve for the ResNet50 architecture

For the second dataset, the architectures tested include EfficientNetB0, Xception, DenseNet121, InceptionV3, VGG16, VGG19, and MobileNet. Among these, the Xception model's accuracy was the greatest of 95.73%, exhibiting exceptional performance in recognizing skin disease patterns. Table III below shows the accuracy of the architectures in the second dataset i.e., "Skin Disease Dataset" and graphically represented in Fig.5. From the figure it could be inferred that Xception Net performed better than other CNN models. It is a light weight model which gives good accuracy in many fields.

TABLE III. ARCHITECTURE PERFORMANCES

CNN Architectures	Accuracy
EfficientNetB0	93.16
Xception	95.73
DenseNet121	92.31
InceptionV3	87.61
VGG16	91
VGG19	90
MOBILENET	67

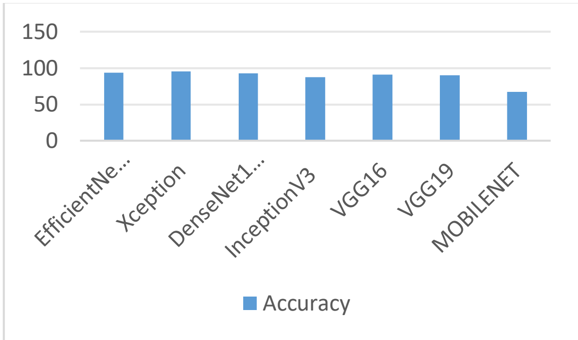

Fig. 5. Performance analysis of CNN models on dataset 2

EfficientNetB0 and DenseNet121 also performed well, with accuracies of 93.16% and 92.31%, respectively, indicating strong classification capabilities. VGG16 and VGG19 achieved accuracies of 91% and 90%, showcasing consistent results but with slightly lower accuracy than the top-performing models. InceptionV3, while widely used in image classification tasks, obtained an accuracy of 87.61%, reflecting a moderate level of performance in this specific context. MobileNet, which is optimized for efficiency, recorded the lowest accuracy of 67%, highlighting a compromise between accuracy and speed in complex classification tasks like skin disease detection. Overall, the results indicate that Xception is the most effective model for this dataset, with EfficientNetB0 and DenseNet121 also proving to be reliable options for accurate and robust skin disease classification.

The confusion matrix shows that, in the second dataset, the Xception CNN model performs well overall, with most predictions aligning along the diagonal, indicating high accuracy for most skin disease classes. However, there are notable confusions in Classes 4 and 5, where samples are misclassified as other classes, suggesting potential overlaps or challenges in distinguishing these diseases. Fig.6 Demonstrates the Confusion Matrix for the Xception architecture.

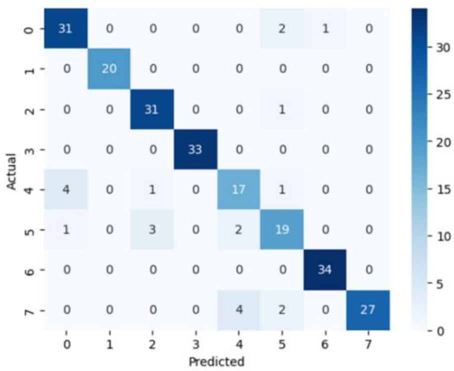

Fig. 6. Confusion Matrix of Xception

The ROC curve for the Xception CNN model on skin disease classification demonstrates excellent performance across all classes, with most areas under the curve (AUC) very close to 1.0. Specifically, classes such as cellulitis, impetigo, nail fungus, chickenpox, and shingles achieve an AUC of 1.0, indicating perfect or near-perfect classification ability for these diseases. Other classes like athlete's foot and ringworm also perform well, with AUC values of 0.99 and 0.98, respectively, showing only minor room for improvement. Fig.7 Demonstrates the ROC curve for the Xception architecture.

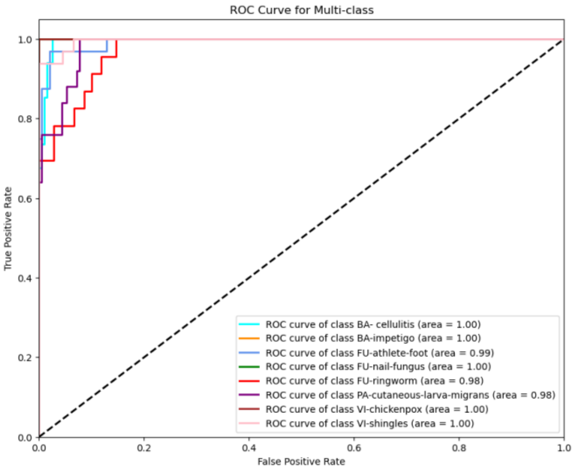

Fig. 7. ROC curve for the Xception architecture

These results highlight the varying effectiveness of CNN architectures in skin disease detection. ResNet50's impressive accuracy of 96.40% is largely due to its residual learning framework, which enables deeper network training while addressing the vanishing gradient problem. Xception, with an accuracy of 95.73%, excels in feature extraction through its depth wise separable convolutions. EfficientNetB0 and DenseNet121 also demonstrated strong performance, achieving accuracies of 93.16% and 92.31%, respectively, with EfficientNetB0 benefiting fsrom its compound scaling. Overall, these results highlight the potential of sophisticated deep learning models, particularly ResNet50 and Xception, to enhance diagnostic precision for skin diseases, ultimately facilitating faster and more effective patient care.

V. Conclusion And Future Work

This study evaluated various architectures of convolutional neural networks (CNNs) for the classification of skin diseases using two distinct datasets. The results indicate that ResNet50 and Xception are among the most effective models, achieving accuracies of 96.40% and 95.73%, respectively. These results highlight the potential of deep learning techniques in enhancing diagnostic accuracy for dermatological conditions. While EfficientNetB3, EfficientNetB0, and DenseNet121 also demonstrated strong performance, other models like VGG16, VGG19, InceptionV3, and MobileNet showed varied capabilities, with MobileNet highlighting the trade-off between model efficiency and classification accuracy. The differences in performance suggest that selecting the appropriate model is crucial for optimizing diagnostic accuracy in clinical settings. Future work may involve exploring ensemble methods that combine multiple architectures to improve classification accuracy and reliability. Additionally, efforts could be made to expandthe dataset to incorporate a wider range of skin conditions and a greater variety of image types, potentially enhancing model generalization. Integrating more advanced Explainable AI (XAI) techniques would also be valuable, as it could increase model transparency, fostering greater trust and adoption in clinical environments. Further optimization through hyperparameter tuning and experimenting with other architectures like DenseNet or Inception could offer additional performance gains in this domain.